Work with real-time data in an Eventhouse in Microsoft Fabric

Overview of Eventhouses and KQL Databases

- **Eventhouse**: A container in Microsoft Fabric that houses one or more KQL databases. It is specifically designed for storing and analyzing large volumes of real-time, continuously arriving data.
- **KQL Databases**: These databases are optimized for **time-series data**, which represents immutable events (events that cannot be changed once they occur).
- **Automatic Partitioning**: Data is automatically organized by **ingestion time**. This allows for rapid access to recent data while maintaining historical data for trend analysis.

Working with Data

- **Ingestion**: You can load data from various sources, including local files, Azure Event Hubs, Fabric Eventstreams, and connectors like Apache Kafka or Amazon Kinesis.
- **OneLake Integration**: You can enable OneLake availability for databases or tables, making the data accessible to other Fabric services like Power BI or Lakehouses.
- **Shortcuts**: You can create database shortcuts to query data from external KQL databases in other eventhouses without copying the data.

Kusto Query Language (KQL)

- **Pipeline Syntax**: KQL uses a "funnel" or pipeline approach where the pipe (|) character flows data from one operator (like where, project, or summarize) to the next.
- **Case Sensitivity**: KQL is strictly case-sensitive for table names, column names, operators, and string values.
- **Query Tools**: Users can use a **KQL Queryset** to run, save, and share queries, or use **Copilot** to generate KQL code using natural language.

Optimization Techniques

To ensure high performance, users should follow several key principles:

- **Filter Early**: Use the where operator (especially time-based filters like ago()) as early as possible in the query to reduce the data scanned.
- **Reduce Columns**: Use project to select only the specific columns needed, reducing processing overhead.
- **Join Strategy**: When joining two tables, always place the **smaller table first** to make the operation more efficient.

Advanced Database Objects

- **Materialized Views**: These are precomputed aggregations that automatically update as new data arrives. They provide instant results for massive datasets by only processing "delta" (new) data rather than recalculating all historical records.
- **Stored Functions**: These allow you to encapsulate a query or transformation logic once and reuse it across multiple queries. They can accept parameters to make repetitive tasks more flexible.

kql

```
TaxiTrips
| where fare_amount > 20
| project trip_id, pickup_datetime, fare_amount
| take 10
```

Questions

1. Your company needs to continuously ingest high-throughput data from an Amazon Kinesis stream into a KQL database. Which approach should be taken?
A. Enable OneLake availability for the KQL database.
B. Manually upload data at regular intervals.
C. Use a connector for non-Microsoft streaming sources.
D. Directly upload local CSV files to the table.

2. What is the primary reason for using stored functions in KQL when analyzing streaming data?
A. To automatically update data streams with new events.
B. To reduce the storage size of databases.
C. To encapsulate repetitive query logic and ensure consistent calculations.
D. To prevent users from viewing the underlying table schema.

3. In KQL, which operator is used to limit the number of rows returned by a query?
A. limit_to
B. restrict
C. take
D. top

4. When designing a KQL query involving joins, why should the smaller table be placed first?
A. It automatically sorts the data, making the join operation faster.
B. It minimizes the number of rows processed during the join operation, improving efficiency.
C. It simplifies the query syntax, making it easier to read and write.
D. The larger table must always be the "right" side of a join per KQL syntax.

5. In a scenario where your team needs to frequently filter and analyze real-time streaming data by specific customer IDs, which feature would best support efficient query reuse?
A. Database shortcuts to access customer data from different sources.
B. Materialized views to store all customer data for quick access.
C. Stored functions that encapsulate filtering logic with customer ID as a parameter.
D. Using the search operator across all tables in the database.

6. Which scenario best illustrates the use of a materialized view in KQL for performance optimization?
A. Joining customer and order tables for a detailed transaction log.
B. Aggregating monthly sales data for a dashboard that updates hourly.
C. Filtering user data based on login timestamps for a one-time report.
D. Creating a shortcut to an external Azure Data Explorer database.

7. Which of the following statements is true regarding the use of materialized views in KQL databases?
A. Materialized views eliminate the need for indexing in KQL databases.
B. Materialized views automatically normalize data for efficient storage.
C. Materialized views reduce query time by maintaining pre-computed results that only need to be updated with new data.
D. Materialized views are only updated when a user manually refreshes them.

8. Your organization wants to create a shortcut to query data from an Azure Data Explorer database within your KQL database in an eventhouse. What feature should you use?
A. Real-Time Dashboards
B. Materialized views
C. Database shortcuts
D. Fabric Activator

9. Which feature would you use to store a common query that can be reused with different parameters in a KQL database?
A. Database shortcuts
B. Materialized views
C. Stored functions
D. KQL Querysets

10. What unique characteristic of real-time data makes automatic time-based partitioning highly effective in Eventhouses?
A. Real-time data is always small in volume.
B. Real-time data represents immutable events tied to specific timestamps.
C. Real-time data is frequently updated after it is written.
D. Real-time data does not require a schema.

11. Which KQL operator is equivalent to the "funnel" concept, where you select specific columns to pass to the next step?
A. where
B. summarize
C. project
D. extend

12. When querying a materialized view, how does the system ensure the results are fresh and up-to-date?
A. It forces a full recalculation of all historical data.
B. It combines the precomputed "materialized part" with the "delta" of new data at query time.
C. It only returns data that was processed in the last background update.
D. It uses a cache that is updated every 24 hours.

13. In KQL, which of the following is true regarding case-sensitivity?
A. Only string values are case-sensitive.
B. Keywords like where are case-sensitive, but table names are not.
C. KQL is case-sensitive for table names, column names, operators, and functions.
D. KQL is completely case-insensitive.

14. What is the benefit of using the ago() function in a where clause?
A. It makes the query run on historical data only.
B. It allows for efficient time-based filtering using the database's time index.
C. It automatically summarizes data into 1-hour buckets.
D. It converts string dates into the proper datetime format.

15. If a team member wants to use natural language to generate a KQL query, which Fabric feature should be enabled by the administrator?
A. Fabric Activator
B. Real-Time Hub
C. Copilot for Real-Time Intelligence
D. KQL Queryset shortcuts

16. What happens to the underlying data when a "database shortcut" is deleted?
A. The underlying data in the source database is deleted.
B. The data is moved into OneLake storage permanently.
C. Only the shortcut is removed; the underlying data remains unaffected.
D. The underlying data is archived for 30 days.

17. Why is "filtering early" considered a key optimization technique in KQL?
A. It reduces the amount of data processed by all subsequent operations in the pipeline.
B. It ensures that the output is automatically sorted.
C. It allows the join operator to work with three or more tables.
D. It is the only way to use the summarize operator.

18. Which of the following can be used to visualize KQL query results directly within Microsoft Fabric?
A. Real-Time Dashboards
B. Power BI reports
C. Built-in charts in a KQL Queryset
D. All of the above

19. What is the primary function of an "Eventhouse" in Microsoft Fabric?
A. To serve as a relational database for transactional ERP systems.
B. To act as a container for KQL databases optimized for real-time data analysis.
C. To replace OneLake as the primary storage for all Fabric items.
D. To manage the scheduling of Spark jobs and notebooks.

Answer

1: C
2: C
3: C
4: B
5: C
6: B
7: C
8: C
9: C
10: B
11: C
12: B
13: C
14: B
15: C
16: C
17: A
18: D
19: B